

Aerodynamic Development of a 2025 Formula 1 Concept Car Using CAD and CFD-Led Iterative Optimisation

Author: Toby Mason

Date of Publication: 29/05/2026

Software / Tools: SolidWorks 2020, AirShaper, LearnSolidWorks.com

Abstract

This project documents the aerodynamic development of a detailed 2025 Formula 1 concept car created in SolidWorks 2020 and analysed using AirShaper. The work was completed as a guided design and CFD project based on LearnSolidWorks.com, but the CAD model itself was personally built in full detail using surface and solid-body modelling methods. The project followed an iterative workflow in which an initial styling model was analysed aerodynamically, design weaknesses were identified, and geometry changes were then introduced across two optimisation cycles. The objective was not only to improve aerodynamic performance, but also to use simulation outputs to guide defensible design changes.

The initial styling model simulation was run in a moving air, on ground setup with rotating wheels at a wind speed of 55.55556 m/s using a steady-state RANS solver with the k-omega SST turbulence model. The styling model returned a drag coefficient of 0.731 and a lift coefficient of -0.497, and a reported total drag force of 2653.70 N and a total vertical force of -1805.53 N.

Following the first simulation review, design changes were made to the front wing, brake ducts, sidepods, floor, fences, top air intake and rear wing. These changes were in response to issues such as flow detachment on the front wing underside, detachment around the brake duct and sidepod inlet, lack of floor downforce, and insufficient vortex generation from the fences.

The first optimisation study increased total downforce to -2702.83 N and changed the full-vehicle coefficients to $C_d = 0.891$ and $C_l = -0.764$. The second optimisation study further increased total downforce to -6175.03 N, with final coefficients of $C_d = 0.893$ and $C_l = -1.673$. The final concept produced substantially greater negative lift than the initial styling model, albeit with increased aerodynamic drag.

Overall, the project demonstrates a simulation-led aerodynamic development process in which CAD refinement, CFD feedback, and iterative design judgement were used together to evolve a concept car from a styling focused baseline into a significantly higher downforce aerodynamic package. The work developed practical capability in complex CAD modelling, aerodynamic interpretation, and structured optimisation in a motorsport context.

Table of Contents

Abstract	1
List of Figures:	2
List of Tables:	3
1. Introduction.....	3
1.1 Formula 1 Aerodynamic Background	5
2. Project Aim	7
3. Project Objectives.....	7
4. Design and Development Workflow.....	7
5. CAD Model Development	7
6. CFD Methodology	10
7. Baseline Styling Model Results	11
8. Design Issues Identified After the Initial CFD Study	12
9. First Optimisation Cycle	12
10. Design Issues Identified After Optimisation 1	14
11. Second Optimisation Cycle / Final Result	14
12. Comparative Performance Summary.....	16
12.1 CFD setup consistency summary	17
13. Discussion.....	18
14. Engineering Learning and Portfolio Relevance	19
15. Limitations	19
16. Conclusion	20

List of Figures:

Figure 1 – Representative 2025-generation Formula 1 aerodynamic architecture	4
Figure 2 – Front three-quarter CAD render of the complete 2025 Formula 1 concept car developed in SolidWorks 2020.	8
Figure 3 – Rear three-quarter CAD render of the complete 2025 Formula 1 concept car developed in SolidWorks 2020	8
Figure 4 – Top view of the complete CAD	9
Figure 5 – Side-profile view of the complete CAD	9
Figure 6 – Styling model in AirShaper.....	11
Figure 7 – Styling model drag clouds in AirShaper.....	11
Figure 8 – Post 1st optimisation model in AirShaper	13
Figure 9 – Post 1st optimisation drag clouds in AirShaper	13
Figure 10 – Post second optimisation/final model in AirShaper	15
Figure 11 – Post 2st optimisation/final drag clouds in AirShaper	15
Figure 12 - Bar chart comparing drag coefficient (Cd) and lift coefficient (Cl)	17
Figure 13 - Bar chart comparing total drag force and total vertical forces	17

List of Tables:

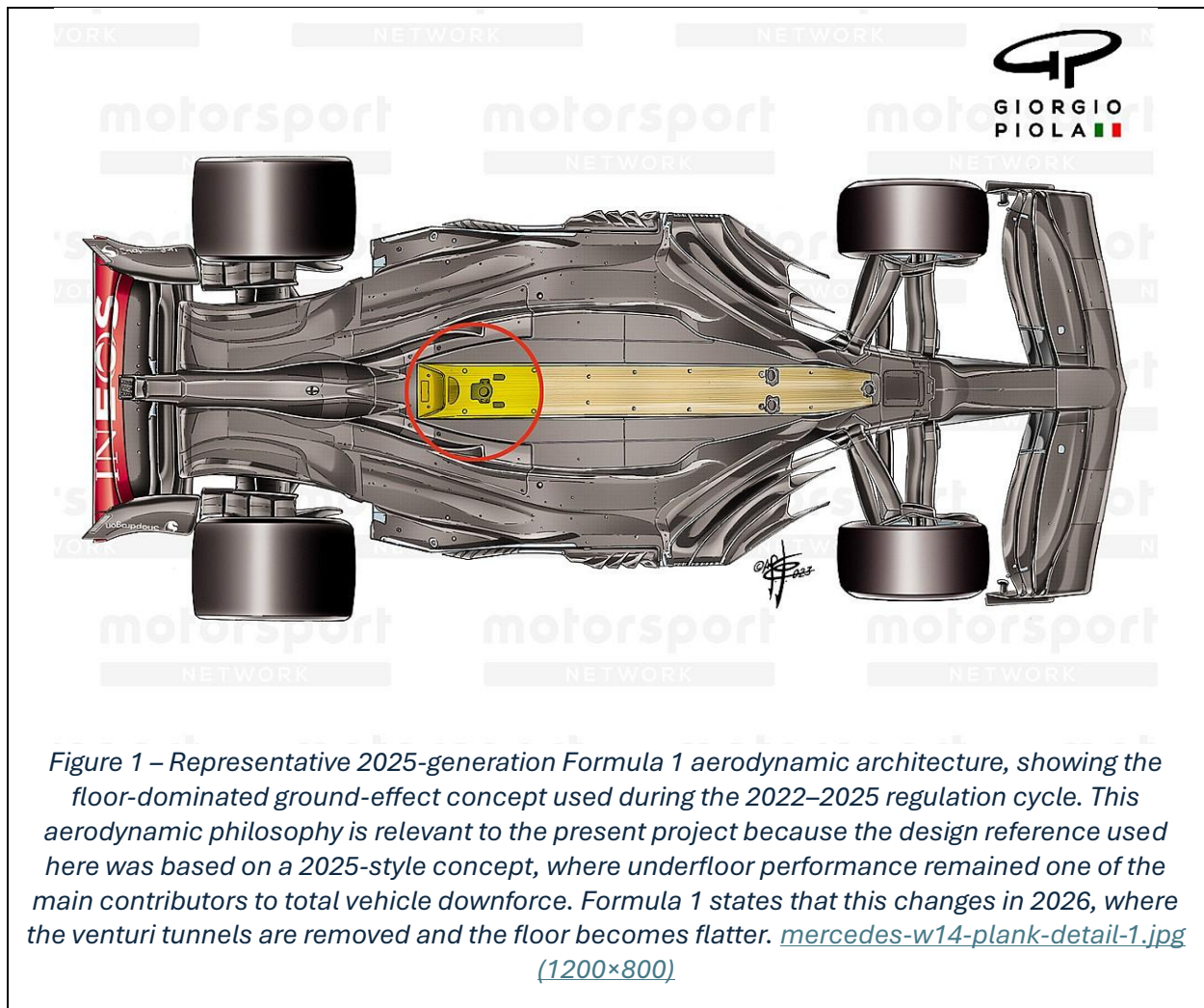
Table 1 - Core Formula 1 aerodynamic concepts relevant to this project	6
Table 2 - CFD setup summary	10
Table 3 - Component-level design change log	16
Table 4 - Performance change by iteration	16

1. Introduction

Aerodynamic performance is one of the dominant performance drivers in modern Formula 1 vehicle design. The bodywork, floor, wings, ducts and ancillary devices do not simply reduce drag or generate downforce independently; they function as an interacting aerodynamic system that influences straight line speed, cornering potential, stability and wake behaviour. For this reason, even concept level design work benefits from structured CFD led evaluation rather than relying on shape alone.

This project was undertaken to strengthen practical capability in motorsport oriented CAD and aerodynamic development. The project was based on a guided workflow from LearnSolidWorks.com, but the full 2025 Formula 1 concept car model was personally designed in SolidWorks 2020 using detailed surface and solid body modelling. The resulting digital model was then analysed in AirShaper and refined through two optimisation loops using the findings from each CFD report as the basis for targeted design changes.

The outcome is therefore not just a rendering exercise or a one-off simulation, but an example of a structured engineering process: create, analyse, diagnose, modify, and re-evaluate.



1.1 Formula 1 Aerodynamic Background

Formula 1 aerodynamic performance is governed primarily by the interaction between downforce and drag. Formula 1 explains that downforce is the vertical aerodynamic force that pushes the car into the track to generate more grip and allow higher cornering speeds, while drag is the resistance that slows the car on the straights. As a result, aerodynamic development is never focused on one quantity alone; the engineering challenge is to generate as much useful vertical load as possible without creating excessive aerodynamic resistance. Also teams alter aerodynamic configuration depending on the circuit, with lower drag set ups favoured at straight line sensitive tracks and higher downforce set ups favoured at corner dominated tracks.

The design reference used in this project was based on the 2025-generation Formula 1 car, which still belonged to a ground effect era. In this generation, the floor was one of the dominant downforce producing devices on the car. The Venturi effect occurs when airflow is accelerated through a narrowed passage, causing a reduction in pressure. In an F1 context, the underfloor tunnels accelerate the airflow beneath the car, creating a low pressure region that helps pull the car toward the track, while the diffuser helps extract and expand that flow downstream. This made the floor a highly influential aerodynamic device in the 2022–2025 regulations. By contrast, Formula 1 states that the 2026 rules remove the ground effect generating venturi tunnels and replace them with a flatter floor and larger diffuser, meaning the newest cars are no longer as floor dominated as the 2025 generation concept used in this project.

Flow behaviour around the car is equally important. When air remains attached to a surface, the device is more likely to produce the intended aerodynamic effect. When the flow detaches or separates, performance generally deteriorates through increased drag, poorer local pressure behaviour, and lower quality flow delivery to downstream devices. This is directly relevant to the present project because the initial optimisation notes identified flow detachment on the front wing underside, brake ducts, and sidepod inlet edges, showing that the early design weaknesses were not only geometric but aerodynamic in nature.

Vortices are another critical aspect of Formula 1 aerodynamics. Vortices as rotating flow structures generated by pressure differentials around aerodynamic surfaces and explains that they can be used as a performance tool rather than being treated purely as a byproduct of the flow. Useful vortices can help guide disturbed flow, improve flow organisation around the car, and help seal sensitive regions such as the underfloor edge. However, vortices also bring risks: if they become too strong, they can add drag, and if they break down, they turn from controlled rotational flow into more chaotic turbulence that harms aerodynamic consistency. This matters in the present project because the floor fences were specifically identified as not creating the intended vortical structures during the baseline design review.

A further concept central to Formula 1 aerodynamics is aerodynamic balance, meaning how the total vertical aerodynamic load is distributed between the front and rear axles. Too much aerodynamic pressure at the rear tends to promote understeer, while too much pressure at the front tends to promote oversteer. This means a car can produce more overall downforce and still feel worse to drive if the front to rear distribution becomes unfavourable. In the present project, the front and rear lift coefficients changed significantly between the styling model, optimisation 1, and optimisation 2, indicating that the iterative geometry changes influenced not only total aerodynamic load, but also the likely handling balance of the car at speed.

Table 1 - Core Formula 1 aerodynamic concepts relevant to this project

Concept	Summary	Relevance to this project
Downforce	Vertical aerodynamic force that pushes the car into the track, increasing tyre grip and cornering performance. Formula 1 notes that wings and the floor generate this load to help the car corner faster.	The optimisation process aimed to increase total downforce, with vertical force changing from -1805.53 N in the styling model to -6175.03 N in the final configuration.
Drag	Aerodynamic resistance opposing forward motion. Lower drag improves straight line speed, while higher drag is often accepted if it delivers useful cornering load.	Drag increased across the iterations, showing the expected trade-off between higher downforce and aerodynamic resistance. Drag force rose from 2653.70 N to 3297.62 N .
Ground effect / Venturi floor	The 2025 generation Formula 1 car used as the basis for this project still relied heavily on venturi-style underfloor tunnels to generate downforce through low pressure beneath the car. Accelerated airflow through the floor tunnels reduces pressure and increases suction, while the diffuser helps extract and sustain that flow.	This is directly relevant because several design changes in the project targeted the floor, rake, and floor intake geometry after poor initial underfloor performance was identified.
2026 aerodynamic shift	Formula 1 states that the 2026 regulations remove the ground-effect venturi tunnels and move to a flatter floor and larger diffuser, meaning the newest cars are less floor-dominated than the 2022–2025 generation.	Useful to mention so the report clearly shows that this project belongs to the 2025 ground-effect generation, not the newer 2026 concept.
Flow attachment and separation	Attached flow follows the intended surface geometry and allows an aerodynamic device to work efficiently. Separated flow creates a thicker wake, increases drag, and can reduce downforce.	Several baseline issues were linked to flow detachment, including the front wing underside, brake duct surfaces, and sidepod inlet edges.
Vortices	Vortices are rotating flow structures created by pressure differences around aerodynamic surfaces. They can be used to guide wake, energise flow, and help seal the underfloor, but if they break down they can increase drag and reduce performance.	This is relevant because the floor fences were specifically identified as not producing the intended vortical behaviour in the early design review.
Aerodynamic balance	Aerodynamic balance is the front to rear distribution of total aerodynamic load. Too much front aero load promotes oversteer, while too much rear aero load promotes understeer.	The front and rear lift coefficients changed noticeably across the three stages, showing that the project was changing not only total load but also the aerodynamic balance of the car.
Understeer / oversteer link	A rear-biased aero balance can contribute to understeer, while an excessively front-biased balance can contribute to rear instability and oversteer.	This helps explain why front/rear load split matters in the reported CFD results, not just the total downforce value.

2. Project Aim

The aim of this project was to develop a detailed 2025 Formula 1 concept car in SolidWorks 2020 and improve its aerodynamic performance through an iterative CFD led optimisation process using AirShaper.

3. Project Objectives

- create a full detailed Formula 1 concept car model using surface and solid body CAD methods
- perform a baseline aerodynamic assessment of the initial styling model
- identify the major sources of drag, flow detachment, weak pressure management, or poor downforce generation
- document and implement targeted geometry changes after each CFD study
- re-run aerodynamic studies after the first optimisation and then after the second optimisation
- compare the resulting drag and vertical-force outputs across all iterations
- demonstrate a structured simulation-led development workflow suitable for use in an engineering portfolio.

4. Design and Development Workflow

The project followed a three stage aerodynamic development route:

Initial styling model creation

A full 2025 Formula 1 concept vehicle was modelled in SolidWorks 2020 using detailed surfaces and solid bodies. The CAD phase focused on producing a complete concept car rather than a single isolated aerodynamic component.

Baseline CFD evaluation

The original styling focused model was assessed in AirShaper to identify the main aerodynamic weaknesses, including local flow detachment, poor floor performance, and ineffective vortex generation in key areas.

Iterative optimisation

Two optimisation loops were then completed. The first was based on the findings from the styling model CFD study, and the second was based on the findings from the first optimisation study. This produced a clear process of analyse → diagnose → modify → re-evaluate rather than a single one-off CFD run.

5. CAD Model Development

The CAD phase focused on creating a full Formula 1 concept car rather than only an isolated aerodynamic device. The vehicle was modelled in detail in SolidWorks 2020, using both surface features and solid bodies to build up the main aerodynamic and structural volumes. This included the primary wings, body contours, sidepod forms, floor region, brake duct areas and upper body surfaces.



Figure 2 – Front three quarter CAD render of the complete 2025 Formula 1 concept car developed in SolidWorks 2020. The image shows the full aerodynamic package layout, including the front wing, sidepods, floor, rear wing, and upper body surfaces, and illustrates the overall design architecture used as the basis for the CFD studies.



Figure 3 – Rear three quarter CAD render of the complete 2025 Formula 1 concept car developed in SolidWorks 2020. This view highlights the rear wing assembly, diffuser region, rear bodywork, and floor geometry, which were key areas in the aerodynamic development process.

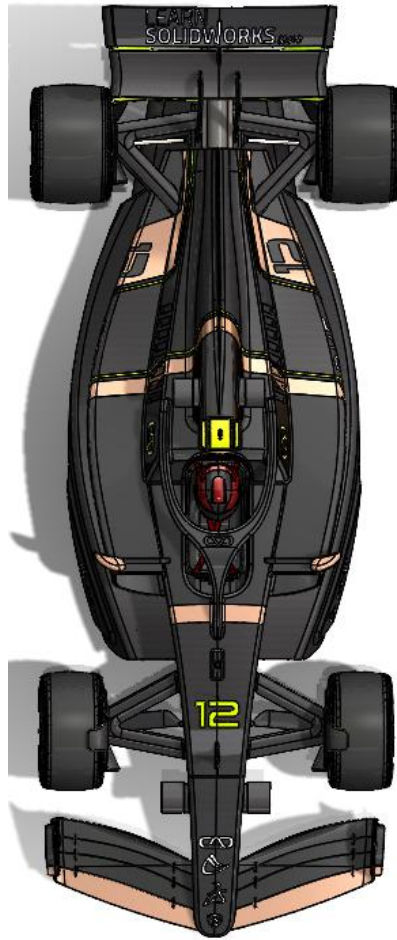


Figure 4 – Top view of the complete CAD assembly, showing the planform layout of the front wing, nose, sidepods, floor edges, rear bodywork, and rear wing assembly.



Figure 5 – Side profile view of the complete CAD assembly, showing the longitudinal body profile, floor rake, sidepod geometry, front wing incidence, and rear aerodynamic packaging.

6. CFD Methodology

All three aerodynamic studies were completed in AirShaper using a moving air, on ground simulation setup with rotating wheels enabled. The fluid was air, the solver type was steady-state RANS, and the turbulence model used was k-omega SST in each case. The simulated wind speed was 55.55556 m/s, with air density set to 1.225 kg/m³, atmospheric pressure set to 101300 Pa, and temperature set to 15 °C.

The styling model study used 10,419,753 cells and reported bounding-box dimensions of 5.59 m × 2.25 m × 1.20 m and a frontal area of 1.91 m².

The first optimisation study used 10,774,461 cells and reported dimensions of 5.47 m × 2.25 m × 1.15 m and a frontal area of 1.86 m².

The second optimisation study used 10,748,198 cells and reported dimensions of 5.45 m × 2.25 m × 1.20 m and a frontal area of 1.94 m².

The CFD reports themselves note that the results are produced in a “virtual wind tunnel” and that simulation remains a simplification of reality due to modelling, discretisation and iteration errors. The reports also explicitly recommend physical wind tunnel testing for higher confidence validation in safety critical applications.

Table 2 - CFD setup summary

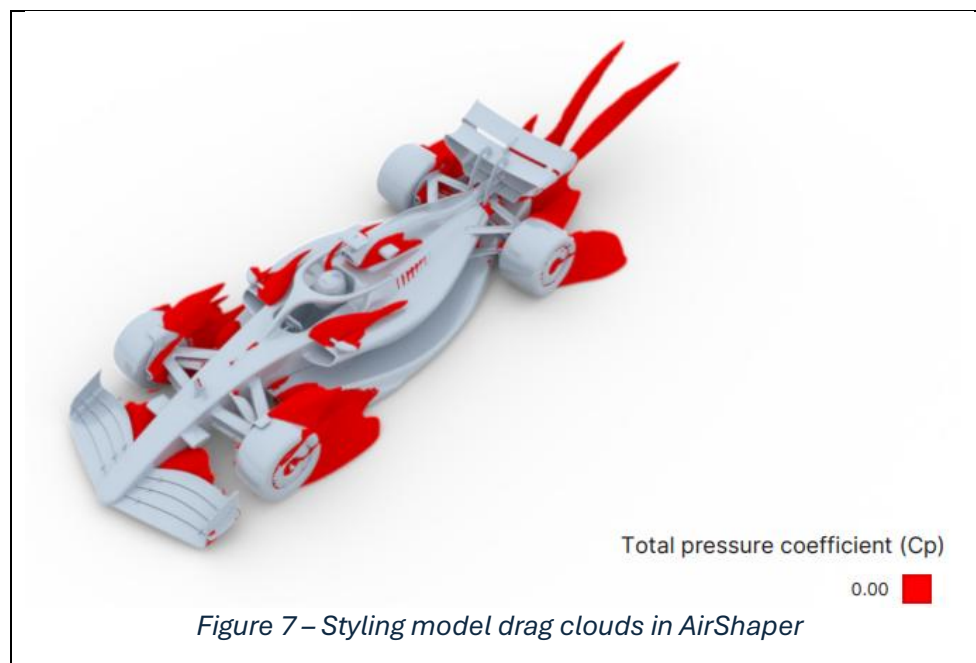
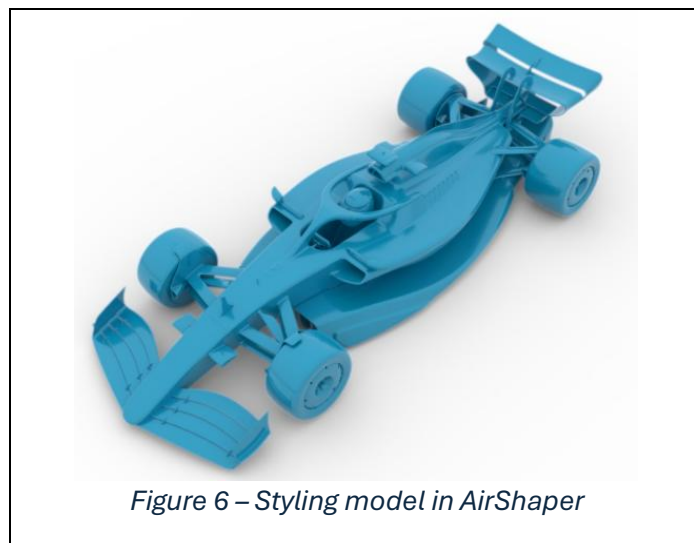
Parameter	Styling model	Optimisation 1	Optimisation 2
Tool	AirShaper		
Solver type	RANS – steady state		
Turbulence model	k-omega SST		
Wind speed	55.55556 m/s		
Rotating wheels	Yes		
Number of cells	10,419,753	10,774,461	10,748,198
Frontal area	1.91 m ²	1.86 m ²	1.94 m ²

Maintaining broadly consistent CFD conditions across the three studies is important because it makes changes in force and coefficient outputs more useful for comparative concept development. The present report should therefore be interpreted as a simulation-led comparison of design states rather than an experimentally validated absolute aerodynamic model.

7. Baseline Styling Model Results

The baseline vehicle was analysed, and the total drag force was 2653.70 N and the total vertical force was -1805.53 N in the wind tunnel coordinate system. The corresponding aerodynamic coefficients were $C_d = 0.731$ and $C_l = -0.497$, with a front lift coefficient of -0.133 and a rear lift coefficient of -0.364. Convergence was reported after 337 iterations.

These values indicate that the styling model already generated negative lift, meaning overall downforce, but the magnitude of the aerodynamic loading was still relatively modest when compared with the later optimised versions. The report also provided drag analysis, pressure mapping, friction mapping, streamline visualisations, 2D flow sections and a noise approximation package, giving a broad basis for identifying where the design could be improved.



8. Design Issues Identified After the Initial CFD Study

The first review of the styling model led to a documented list of improvement areas. The file explicitly identified the following issues and intended changes:

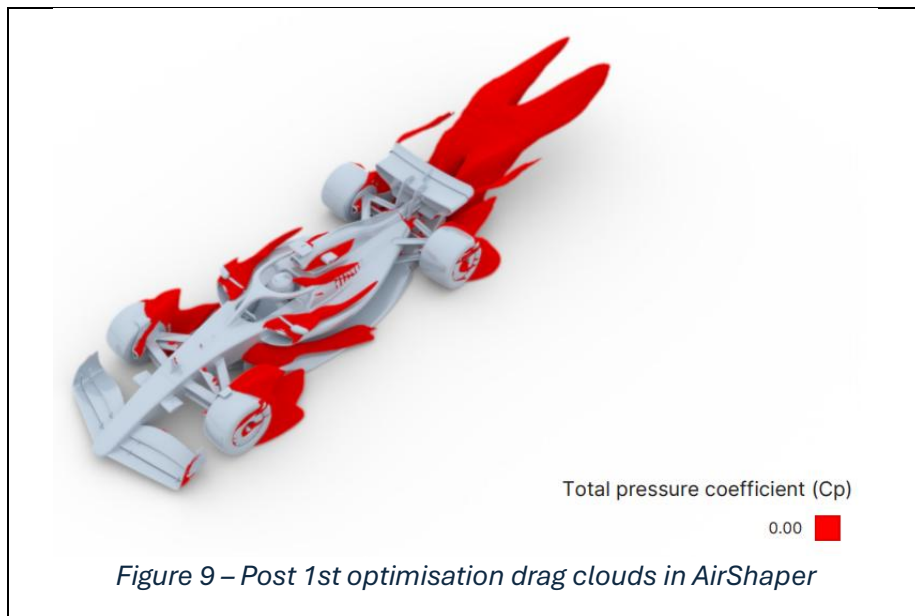
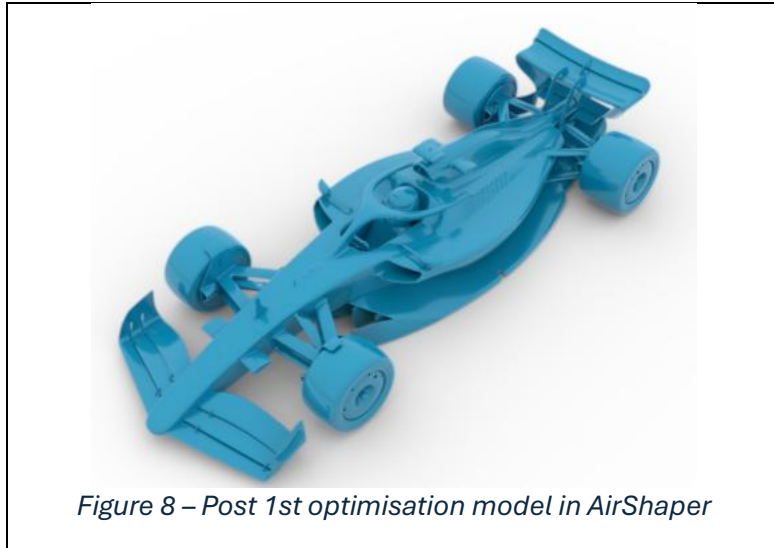
- **Front wing:** flow detachment on the underside of the aerofoils; add fillets to sharp corners; align the front wing aerofoils; change aerofoil profiles to a more tear-drop form.
- **Brake duct:** flow detachment; round the duct.
- **Sidepod:** detachment from the sharp edge of the sidepod inlet; introduce curved geometry and larger fillets; reduce mirror support bar thickness.
- **Floor:** generating no downforce and lacking venturi effect; adjust air-intake size and sharpness; decrease rear rake angle. Also, a low pressure area behind the tyres pulling air into the surrounding region and recommends revising floor flange geometry.
- **Fences:** not creating vortices; use larger fences towards the floor edge.
- **Top air intake:** flow stagnation; apply large edge fillets and bring the surface further forward.
- **Rear wing:** last aerofoil creating no downforce; improve aerofoil geometry and move the drag aerofoil more vertical.

9. First Optimisation Cycle

The first optimised configuration was analysed and this model produced a total drag force of 3151.47 N and a total vertical force of -2702.83 N. The aerodynamic coefficients changed to $C_d = 0.891$ and $C_l = -0.764$. The front lift coefficient became +0.063, while the rear lift coefficient was -0.827. Convergence was detected after 594 iterations.

Relative to the styling model, the first optimisation stage therefore produced a clear increase in total negative lift/downforce magnitude, moving from -1805.53 N to -2702.83 N, while drag also increased from 2653.70 N to 3151.47 N.

This first optimisation pass appears to have shifted the aerodynamic balance substantially toward the rear, as the reported front lift coefficient was positive (+0.063) while the rear lift coefficient remained strongly negative (-0.827).



10. Design Issues Identified After Optimisation 1

After the first optimisation cycle, improvement was recorded for a second set of modifications for the next development stage:

- **Front wing:** the endplate had become straight, leading to drag from leading-edge detachment; re-add the outward draft present on the first model.
- **Front wing canard:** producing no downforce; convert the feature to a more aerofoil like shape.
- **Brake duct:** top edge still too sharp and still causing flow detachment; curve the top edge.
- **Sidepods:** the sidepod angle was too large following rake reduction; bring the sidepod into a more horizontal orientation.
- **Floor:** round the sharp air intake edge more; raise rear rake and make the rear floor walls lower because the front/mid-floor was too low and air was stalling, increasing pressure.
- **Rear wing:** adjust the shape and angle of attack of the beam wing.

11. Second Optimisation Cycle / Final Result

The final optimised model was analysed, this study reported a total drag force of 3297.62 N and a total vertical force of -6175.03 N. The final aerodynamic coefficients were $C_d = 0.893$ and $C_l = -1.673$, with a front lift coefficient of -0.459 and a rear lift coefficient of -1.214. Convergence was detected after 3283 iterations.

Compared with the initial styling model, the final optimisation stage produced a much larger total negative lift magnitude, moving from -1805.53 N to -6175.03 N. C_d also increased from 0.731 to 0.893 across the same development path.

Compared with the first optimisation stage, the second optimisation stage retained almost the same drag coefficient level numerically (0.891 to 0.893) but produced a much more negative overall lift coefficient (-0.764 to -1.673). The front coefficient also moved from +0.063 in optimisation 1 to -0.459 in optimisation 2, indicating that the final round of geometry changes recovered front-end downforce in the reported results.

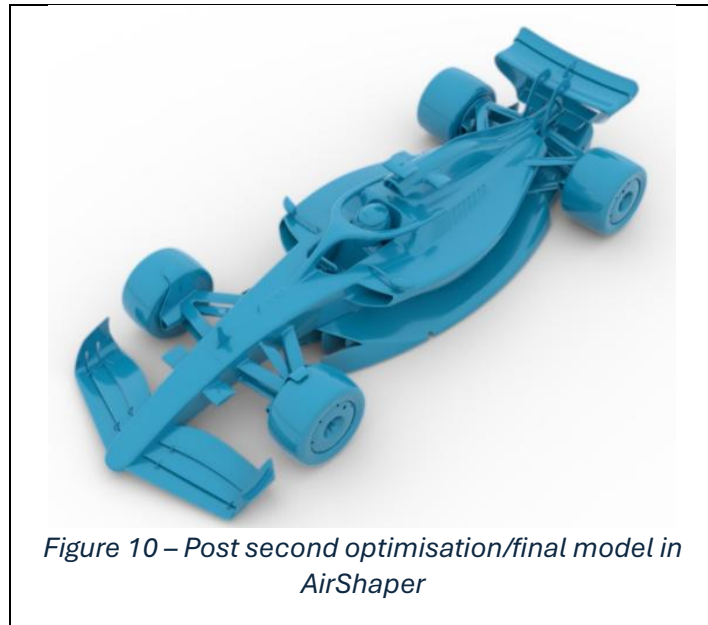


Table 3 - Component-level design change log

Component	Issue identified	Intended change
Front wing	Flow detachment on underside of aerofoils	Add fillets, align aerofoils, and revise aerofoil profiles toward a more tear-drop-like form.
Brake duct	Flow detachment	Round the duct geometry.
Sidepod	Detachment at the sharp air-inlet edge	Introduce curved geometry, larger fillets, and reduce mirror-support thickness.
Floor	No downforce / no venturi effect / poor local flow behaviour	Revise intake size and sharpness, reduce rear rake, and modify floor flange geometry.
Fences	Not creating vortices	Increase fence size toward the floor edge.
Top air intake	Flow stagnation	Add large edge fillets and move the surface further forward.
Rear wing	Last aerofoil creating no downforce	Improve aerofoil geometry and move the drag aerofoil more vertical.

12. Comparative Performance Summary

Table 4 - Performance change by iteration

Metric	Styling model	Optimisation 1	Optimisation 2
Drag force (N)	2653.70	3151.47	3297.62
Vertical force (N)	-1805.53	-2702.83	-6175.03
Drag coefficient, Cd	0.731	0.891	0.893
Lift coefficient, Cl	-0.497	-0.764	-1.673
Front Cl	-0.133	0.063	-0.459
Rear Cl	-0.364	-0.827	-1.214

The comparison shows that the final optimisation stage increased the magnitude of total vertical force by 4369.50 N relative to the styling model, while drag force increased by 643.92 N over the same development path. The numerical trend therefore supports the conclusion that the project evolved from a styling led baseline into a substantially higher downforce aerodynamic configuration, albeit with a clear drag penalty. The front and rear lift coefficient changes also show that the development process affected not only total aerodynamic load but also the front to rear distribution of that load.

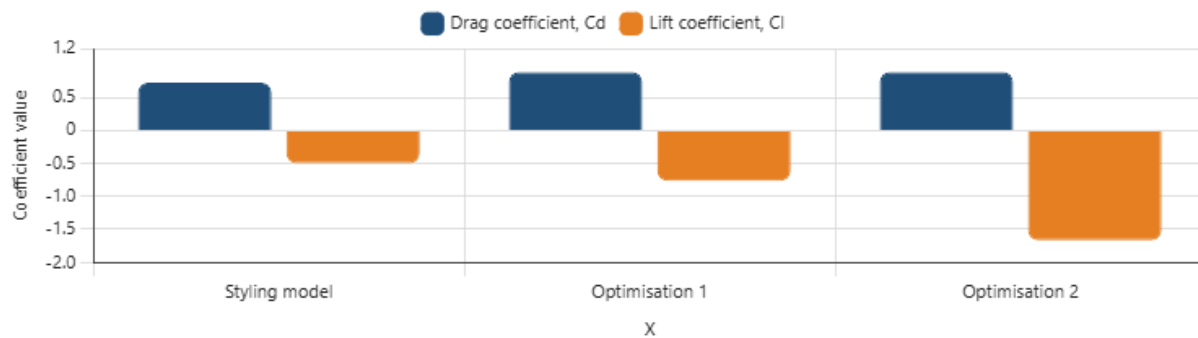


Figure 12 - Bar chart comparing drag coefficient (Cd) and lift coefficient (Cl) across the styling model, optimisation 1, and optimisation 2 CFD studies.

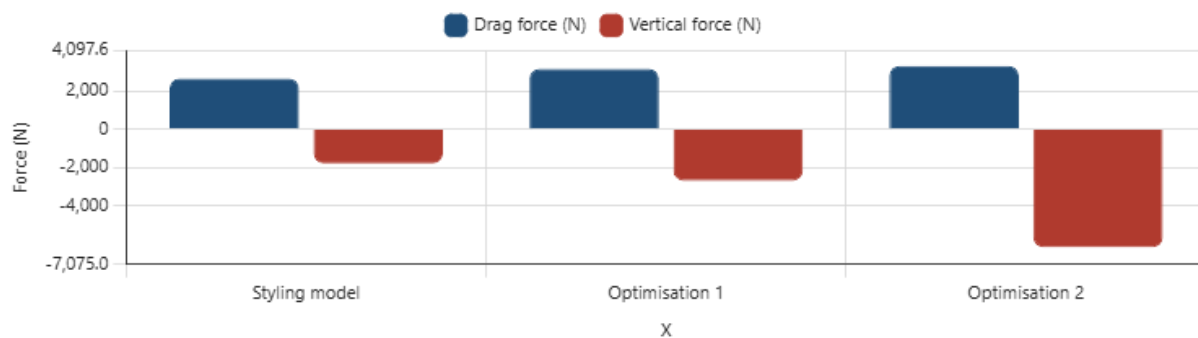


Figure 13 - Bar chart comparing total drag force and total vertical force across the styling model, optimisation 1, and optimisation 2 CFD studies.

12.1 CFD setup consistency summary

All three studies were run at 55.55556 m/s using air at 15 °C and 1.225 kg/m³ in a steady-state RANS simulation with the k-omega SST turbulence model and rotating wheels enabled.

13. Discussion

The most important outcome of this project is not simply that the final concept produced more downforce than the baseline model, but that the aerodynamic development followed a traceable engineering process. The styling model CFD study identified specific performance weaknesses, those weaknesses were translated into geometry changes, and the revised concept was then re-evaluated through further CFD work. This is a meaningful distinction, because it shows that the project was not approached as a one-off simulation exercise, but as a structured aerodynamic development loop.

The design development path suggests that the concept gradually moved from a styling led shape toward a more aerodynamically purposeful configuration. The baseline review identified recurring issues linked to flow detachment, weak floor behaviour, ineffective vortex generation, and underperforming wing elements. The selected responses were largely geometric in nature: adding fillets, smoothing transitions, reshaping aerofoils, adjusting sidepod and floor geometry, and altering rear-wing configuration, but the purpose of these changes was aerodynamic rather than cosmetic. In other words, the optimisation path was driven by identifiable flow quality problems rather than arbitrary shape changes.

The first optimisation stage increased total aerodynamic load but also shifted the reported aerodynamic balance rearward, with the front lift coefficient becoming positive while the rear remained strongly negative. The second optimisation stage then recovered front end loading, with the front lift coefficient moving back to a negative value and the total lift coefficient becoming much more negative overall. This matters because the usefulness of an aerodynamic package depends not only on the total amount of vertical load generated, but also on where that load acts. Excessive rear aerodynamic bias tends to promote understeer, while excessive front bias tends to promote oversteer. The reported coefficient progression therefore suggests that the second optimisation stage improved not only total aerodynamic load, but also the front to rear useability of that load.

From a motorsport engineering perspective, this is valuable because it demonstrates an understanding that aerodynamic development is a systems problem. Local flow separation on the front wing, brake duct, sidepod, or floor edge does not remain a purely local issue; it changes the quality of the airflow delivered to downstream devices and can reduce the effectiveness of the whole-car aerodynamic package. The same principle applies to vortical structures. Vortices can be highly useful when controlled because they help organise flow and preserve the effectiveness of critical regions such as the underfloor, but they may also create drag or degrade into less useful turbulence if they become unstable. This broader systems view is what gives the project more technical value than a simple before-and-after coefficient comparison.

At the same time, the project clearly demonstrates the central aerodynamic trade-off between downforce and drag. The final optimisation stage delivered a significantly larger downforce state than the styling model, but it also increased drag. This means the final concept should not be described as universally superior in every aerodynamic sense; instead, it should be described more accurately as a substantially higher-downforce configuration developed through iterative CFD-led refinement. That is the more technically correct interpretation, and it reflects the same design compromise faced in real motorsport aerodynamic development.

14. Engineering Learning and Portfolio Relevance

This project developed capability in several areas directly relevant to motorsport engineering roles:

- Detailed CAD construction of a full Formula 1 concept vehicle using surfaces and solids.
- Interpretation of integrated full vehicle aerodynamic outputs rather than isolated component only results.
- Reading CFD outputs such as force balance, coefficient changes, convergence behaviour and flow-visualisation packages.
- Turning simulation observations into documented geometry changes across multiple subsystems.
- Understanding that aerodynamic development is an iterative process involving trade-offs rather than a single “correct” design.

15. Limitations

The AirShaper reports explicitly state that the results are produced using a “virtual wind tunnel” and that the simulations remain a simplification of reality because of modelling, discretisation, and iteration errors. The reports also recommend physical wind tunnel testing for higher confidence validation when greater accuracy is required.

The present study should therefore be interpreted as a comparative aerodynamic development exercise rather than an experimentally validated full performance model. The aerodynamic trends and relative changes between design states are useful for concept evaluation and geometry refinement, but the results should not be treated as direct substitutes for physical test correlation. Additional practical limitations include the absence of experimental validation and the fact that the report evidence is based primarily on the CFD outputs and the documented design-change notes.

16. Conclusion

This project demonstrated the aerodynamic development of a detailed 2025 Formula 1 concept car through an iterative simulation led workflow. A complete concept vehicle was created in SolidWorks 2020, analysed using AirShaper, and progressively refined through two optimisation cycles informed by CFD findings and documented design changes.

The reported results show a clear shift from a styling-led baseline toward a substantially higher-downforce aerodynamic package. The styling-model study reported **Cd = 0.731** and **Cl = -0.497**, while the final optimisation study reported **Cd = 0.893** and **Cl = -1.673**. Total vertical force changed from **-1805.53 N** in the baseline study to **-6175.03 N** in the final configuration. These figures show that the development process produced a major increase in aerodynamic loading, although with a corresponding increase in drag.

More importantly, the project demonstrates the engineering logic behind aerodynamic development: use simulation to identify weak regions, translate those findings into geometry changes, and reassess the resulting concept through further analysis. That workflow is directly relevant to motorsport engineering because it reflects the iterative relationship between CAD, CFD, aerodynamic interpretation, and design decision-making. As a portfolio project, the report therefore demonstrates not only modelling and analysis capability, but also a process-based understanding of how aerodynamic performance is developed and improved.

- END OF DOCUMENT -